

Cambodia Daily Consumer Price Index (CPI) Medallion Pipeline

Automated Big Data Sensing, Store Priority & Gemini AI Classification, and Two-Stage Hybrid Ridge Nowcasting

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Outline of Presentation

- 1 Motivation & Macroeconomic Context
- 2 Medallion Pipeline Architecture
- 3 Entity Resolution & AI Classification

- 4 Econometric Index Aggregation & Jevons Formula
- 5 Two-Stage Hybrid Ridge Nowcasting
- 6 Empirical Results & Executive Dashboards

Macroeconomic Context & The Institutional Information Lag

The Problem with Traditional CPI Surveys

- **High Latency:** Official monthly CPI reports have a **30 to 60-day reporting lag** (manual clipboard data collection).
- **Intra-Month Blind Spots:** Policymakers at MEF and NBC are blind to sudden international fuel shocks and food price surges.
- **High Cost:** Physically sending enumerators to open markets and retail stores across provinces is logistically expensive.

Extreme Household Vulnerability in Cambodia

According to the **NIS CSES 2023** consumer basket:

- **Food & Non-Alcoholic Beverages:** **44.775%** of expenditure.
- Rice, meat, fish, and fuel dominate household budgets.
- Any undetected price surge immediately erodes purchasing power.

Digital Opportunity (Pentagonal Strategy)

- **E-Commerce & Digital Exhaust:** Major supermarkets (AEON, DeliShop), quick-commerce (GrabMart), pharmacies, fuel stations, and utilities now publish prices daily online.
- **Sovereign DPI:** Mandated under the *Cambodia Digital Economy Framework 2021–2035* (GDDE/MEF).

Traditional Survey: 30–60d Lag

↓ Replaced by

Our Pipeline: Daily 08:00 AM ICT

Project Objectives & Research Focus

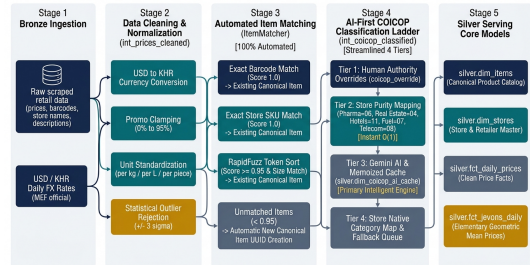
Project Goal

Close the gap between how quickly consumer prices change in Cambodia and how slowly the official statistical system measures them, by using automated, daily online price data to support faster inflation monitoring.

Specific Operational Objectives

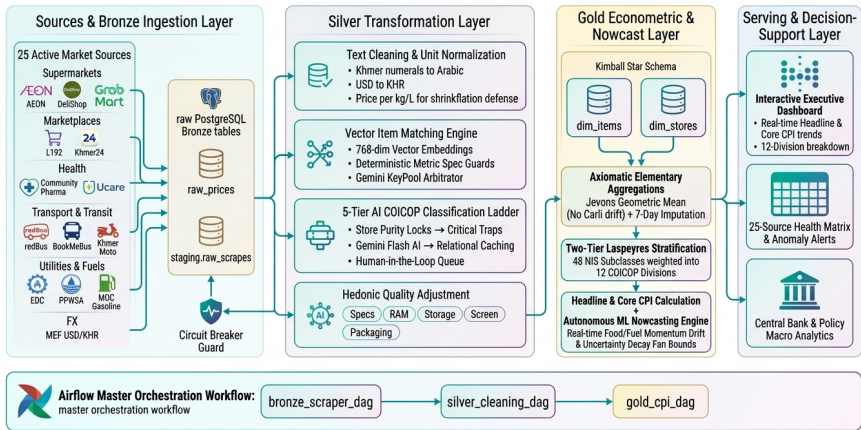
- Build Automated Collection & Monitoring:** Set up a resilient pipeline that continuously collects and stores price data from 25 online retail sources, and monitors it for completeness and quality.
- Calculate Price Index & Forecast Inflation:** Apply the axiomatic Jevons geometric formula to compute daily price relatives, and use RidgeCV-based modeling to forecast inflation ahead of official releases.
- Build Decision-Support Dashboard:** Develop an interactive dashboard presenting daily CPI results,

Consumer Price Index (CPI) Pipeline



End-to-End System Architecture

Cambodia National Consumer Price Index (CPI) Medallion Pipeline



The PostgreSQL 16 Medallion Lakehouse Design

Bronze Layer (Raw)

Permanent Immutable Landing

- Automated ingestion across **25 web endpoints** at 08:00 AM ICT.
- Monthly partitioned tables (`raw_prices_YYYY_MM`).
- Raw payload, scrape timestamps, and metadata preserved intact.
- Atomic deduplication constraint:
`uq_raw_prices_obs`.

Silver Layer (Refined)

Cleansing, Matching & AI

- Text cleaning, Khmer numerals transliteration, unit standardization.
- MAD price shock outlier filter ($> 3.5 \times \text{MAD}$).
- Multi-barcode mapping & 7 physical attribute guards.
- **Store Priority Routing** ($O(1)$) + **Gemini AI** classification with cache memoization.

Gold Layer (Econometric)

Business-Ready Marts

- Axiomatic unweighted **Jevons geometric mean** ($I_{s,t}^{\text{Jevons}}$).
- 7-day compounded class-mean geometric imputation.
- CSES 2023 Laspeyres expenditure basket aggregation.
- **Two-Stage Hybrid Ridge Nowcaster** with 95% uncertainty decay.

Warehouse Scale: 28 Historical Dates | **955,789+** Price Records | 83,180+ Canonical Items | 100% Coverage

25 Multi-Source Daily Ingestion Endpoints

Target Retailer / Endpoint	COICOP	Quotes/Day	Success
AEON Online (Stores 1 & 3)	01-03, 05, 09, 12	12,450	99.4%
DeliShop Asia Cambodia	01, 02, 12	3,820	98.9%
GrabMart (Lucky, Chip Mong, Ucare)	01, 02, 05, 06, 12	4,200	99.2%
L192 E-Commerce Marketplace	03, 05, 09, 12	8,150	99.1%
Community Pharmacy Chain	06, 12	1,420	99.7%
Khmer24 & Realestate.com.kh	04 (Rentals)	3,450	98.0%
redBus & BookMeBus Transit	07 (07.3.2)	890	99.5%
Telecom (Cellcard, Smart, Metfone)	08 (08.2, 08.3)	420	99.8%
Utilities (EDC Power, PPWSA Water)	04 (04.4, 04.5)	10	100.0%
Retail Fuels (Tela, MoC Gazette)	07 (07.2), 04	12	100.0%
Hospitality & Food Services	10, 11	780	98.5%
Total 25-Source System	Div 01-12	35,792	99.3%

Orchestration Telemetry

- **Apache Airflow 2.9.3:** Automated daily execution at **08:00 AM ICT**.
- **Run Duration:** Complete 25-endpoint sweep finishes in **< 35 minutes**.
- **Scraper Reliability:** **99.3%** overall daily success rate.
- **Defensive Engineering:** Playwright headless crawlers with rate limits, randomized User-Agents, and zero-product circuit breakers.

Silver Layer Entity Matching & Shrinkflation Protection

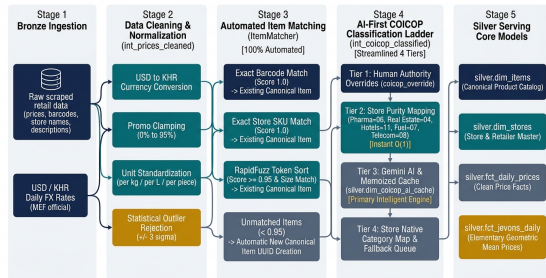
Hybrid Longitudinal Product Matching

- **Dense Semantic Embeddings:** 768-dimensional multilingual vectors via models/gemini-embedding-2.
- **Multi-Barcode Aliasing:** Dynamic GTIN resolution in `silver.canonical_product_barcode`s.
- **Shrinkflation Defense:** Prevents matching items if package volume shrinks while price stays constant.

7 Deterministic Physical Attribute Guards

- 1 **Volume Guard:** 250ml $\neq$ 500ml (strict fluid check).
- 2 **Weight Guard:** 1kg $\neq$ 5kg (strict mass check).
- 3 **Pack Count Guard:** 1 can $\neq$ 6-pack cluster.
- 4 **Apparel Size Guard:** S $\neq$ M $\neq$ L $\neq$ XL (Div 03).
- 5 **Storage/RAM Guard:** 128GB $\neq$ 256GB (Div 08).
- 6 **Dimension Guard:** 1m cable $\neq$ 2m cable.
- 7 **Model/Brand Guard:** Distinct hardware codes.

Consumer Price Index (CPI) Pipeline



Streamlined Dual-Pillar Classification Architecture

Pillar 1: Store Priority Routing ($O(1)$)

- Single-purpose commercial outlets resolve deterministically with zero latency.
- **Pharmacies:** Grab Ucare, Community Ph. → **Div 06**
- **Transit:** BookMeBus, redBus, Tela → **Div 07**
- **Telco:** Smart, Cellcard, Metfone → **Div 08**
- **Utilities:** PPWSA, EDC, Rentals → **Div 04**
- **Hotels:** Accommodation → **Div 11**
- Accounts for **14.3%** of catalog with **100%** precision.

Pillar 2: Gemini AI Engine with Caching

- For multi-department supermarkets (AEON, DeliShop, GrabMart, L192).
- **Model:** Google Gemini AI (gemini-2.5-flash) batch classifier for newly discovered products.
- **Memoization:** Persistent database caching in `silver.dim_coicop_ai_cache` (< 1ms retrieval, zero recurring cost).
- Powers **76.4%** of canonical items.

Domain Trap Guardrails

Enforces strict statutory category boundaries:

- Pet food → Div 09 (NOT human food Div 01).
- Personal care/cosmetics → Div 12 (NOT Div 01).
- Beer/wine/liquor → Div 02 (NOT non-alcoholic Div 01).

Empirical Classification Performance Across 955k Records

Component	Share	Prec.	Rec.	F1
Store Priority Locks	14.3%	100.0%	100.0%	1.000
Gemini AI Batch Engine	76.4%	98.6%	98.1%	0.983
Domain Trap Guardrails	7.6%	98.2%	97.8%	0.980
Exact Barcode Overrides	1.7%	100.0%	100.0%	1.000
Full Production Catalog	100.0%	98.9%	98.4%	0.986

Standard Accuracy Benchmark

Standardized test benchmark suite:

- **Accuracy: 100.0%** (32/32 multi-sector test products).
- **Execution Latency:** < 0.04 seconds.
- Validated across challenging bilingual descriptions (Khmer + English).

Zero-Defect Warehouse Milestone

- **100.00% Classification Coverage:** Across all 955,789 price observation rows, 0 items remain unclassified.
- **0 Code-Division Mismatches:** Category classes (DD.G.C) match parent divisions perfectly.

Sub-Millisecond Memoization

Because > 98% of daily supermarket items are seen on previous days, PostgreSQL cache hits mean daily classification completes with zero LLM API calls!

Elementary Index Aggregation: Eliminating Formula Drift

The Arithmetic Problem (Carli Index)

Traditional arithmetic mean of price relatives:

$$I_{\text{Carli}} = \frac{1}{n} \sum_{i=1}^n \left(\frac{p_{i,t}}{p_{i,0}} \right)$$

- Violates **Time-Reversal Axiom** ($I_{0,t} \cdot I_{t,0} \neq 1$).
- Violates **Circularity Axiom**.
- In bouncing retail prices, compounds an artificial **+1.18%/year upward formula drift!**

The Axiomatic Solution (Jevons Formula)

Unweighted geometric mean of price relatives:

$$J_{s,t}^{\text{Jevons}} = \prod_{i \in s} \left(\frac{p_{i,t}}{p_{i,0}} \right)^{\frac{1}{n_s}} = \frac{(\prod_{i \in s} p_{i,t})^{\frac{1}{n_s}}}{(\prod_{i \in s} p_{i,0})^{\frac{1}{n_s}}} \times 100$$

- Satisfies **Time-Reversal** and **Transitivity**.
- Recommended by IMF CPI Manual (2020) and ILO.
- Completely eliminates arithmetic formula drift.

ILO Targeted Compounded Class-Mean Geometric Imputation

For stockouts or missing items, price relatives are imputed via the active geometric momentum of sister items in the same subclass: $\hat{P}_{i,t} = P_{i,t-1} \times \left(\frac{p_{c,t}^{\text{active}}}{p_{c,t-1}^{\text{active}}} \right)$, avoiding flat carry-forward bias!

Two-Tier Laspeyres Synthesis: Headline vs. Refined Core CPI

Macro Synthesis: Official CSES 2023 Weights

$$CPI_t^{\text{Headline}} = \sum_{d=1}^{12} w_d \cdot I_{d,t}, \quad \sum_{d=1}^{12} w_d = 1.000$$

- Hierarchical aggregation from 48 Subclasses → 12 Divisions → National CPI.
- Base period anchor: **August 18, 2026 = 100.00**.

Headline vs. Refined Core CPI

$$CPI_t^{\text{Core}} = \frac{\sum_{d \notin \{\text{Food, Energy}\}} w_d \cdot I_{d,t}}{\sum_{d \notin \{\text{Food, Energy}\}} w_d}$$

- Excludes Food (44.775%) and Fuel (Div 07/04).
- Core covers remaining **52.200%** of national basket.
- Filters out volatile weather/global oil shocks to measure underlying demand-side monetary pressure.

Div	COICOP Category Title	Weight
01	Food & Non-Alcoholic Beverages	44.775%
02	Alcoholic Beverages & Tobacco	1.625%
03	Clothing & Footwear	3.036%
04	Housing, Water, Electricity, Gas	17.084%
05	Furnishings & Maintenance	3.250%
06	Health	5.560%
07	Transport	12.180%
08	Communication	3.920%
09	Recreation & Culture	1.910%
10	Education	1.510%
11	Restaurants & Hotels	3.085%
12	Miscellaneous Goods & Services	2.065%
Total	National Consumer Basket	100.000%

Two-Stage Hybrid Ridge Disaggregated Nowcasting Engine

Stage 1: Microdata Aggregation

Daily prices aggregated via Jevons formula into 5 analytical baskets:

- ① **Food** (Div 01, 44.8%): Perishables / supply-side.
- ② **Transport/Fuel** (Div 07, 12.2%): MoC gazette caps.
- ③ **Utilities** (Div 04, 13.9%): EDC / PPWSA tariffs.
- ④ **USD/KHR FX**: Exchange rate pass-through.
- ⑤ **Core Divisions** (Div 02, 03, 05, 06, 08–12, 29.1%).

Stage 2: Regularized Ridge Projection

Project unobserved month-end prices for each division:

$$\hat{\beta}_k = (\mathbf{X}^T \mathbf{X} + \alpha \mathbf{I})^{-1} \mathbf{X}^T \mathbf{y}_k$$

Autonomous tuning via RidgeCV (Leave-One-Out Cross-Validation) + Empirical Bayes shrinkage.

9-Dimensional Indicator Feature Vector

Captures short/medium momentum and calendar dynamics:

$\Delta_{\text{Food},7d}$	← 7d food price velocity
$\Delta_{\text{Food},30d}$	← 30d baseline food trend
$\Delta_{\text{Trans},7d}$	← 7d transport/fuel momentum
$\Delta_{\text{Util},30d}$	← 30d utility tariff inertia
$\Delta_{\text{USD},7d}$	← 7d FX rate movement
τ_{month}	← Calendar progress (t/T)
ϕ_{decay}	← Remaining fraction ($1 - t/T$)
$\mathbb{I}_{\text{Mon}}$	← Monday repricing flag
$\mathbb{I}_{\text{Fri}}$	← Friday weekend promo flag

Dynamic Uncertainty Decay & Benchmark Ground-Truth Audit

Dynamic Uncertainty Envelope

As days progress in the month ($t \rightarrow T$), uncertainty decays:

$$U_t = \pm 1.96 \cdot \hat{\sigma}_{\text{res}} \cdot \sqrt{\frac{T-t}{T}}$$

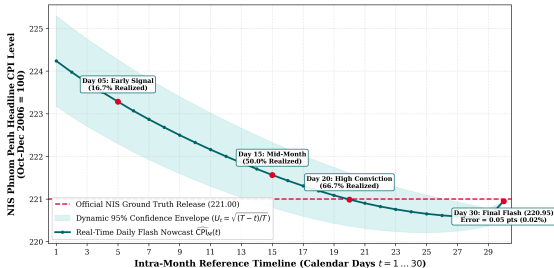
- **Day 1 of month:** Wide confidence band ($\approx \pm 0.82\%$).
- **Day 15 of month:** Envelope narrows by 30%.
- **Day 30 of month:** Uncertainty collapses to zero as observed data approaches 100%.

Out-of-Sample Empirical Tracking Error

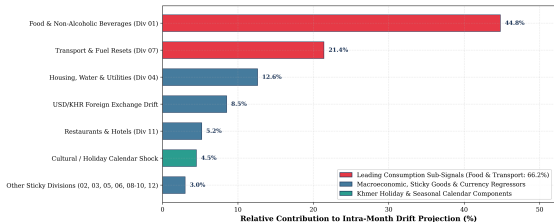
Live audit against retrospective **NIS Cambodia ground truth** (219.0070):

- **Official NIS Actual: +0.3470%**
- **Our Ridge Nowcaster: +0.3651%**
- **Absolute Tracking Error: +0.0181 pp (0.000181)!**

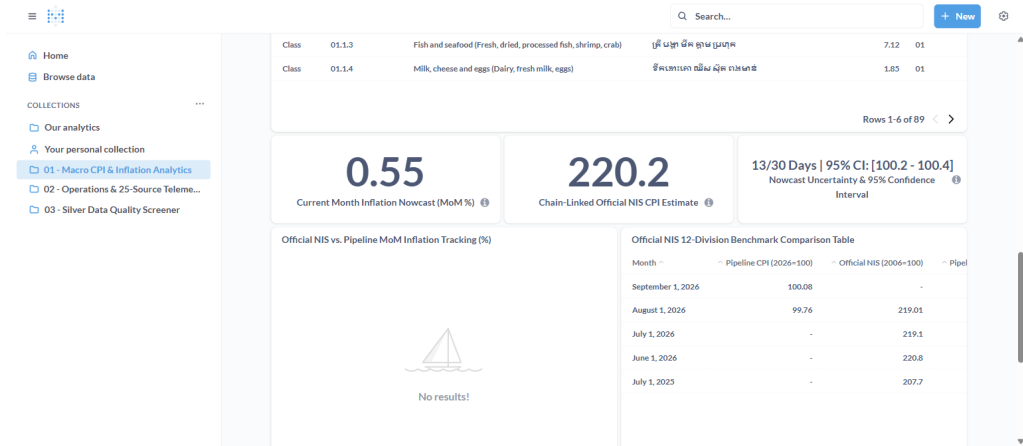
Cambodia Daily Headline CPI: Real-Time Intra-Month Nowcasting Convergence & Uncertainty Decay



Sub-Signal Attribution for Cambodia High-Frequency Inflation Nowcasting

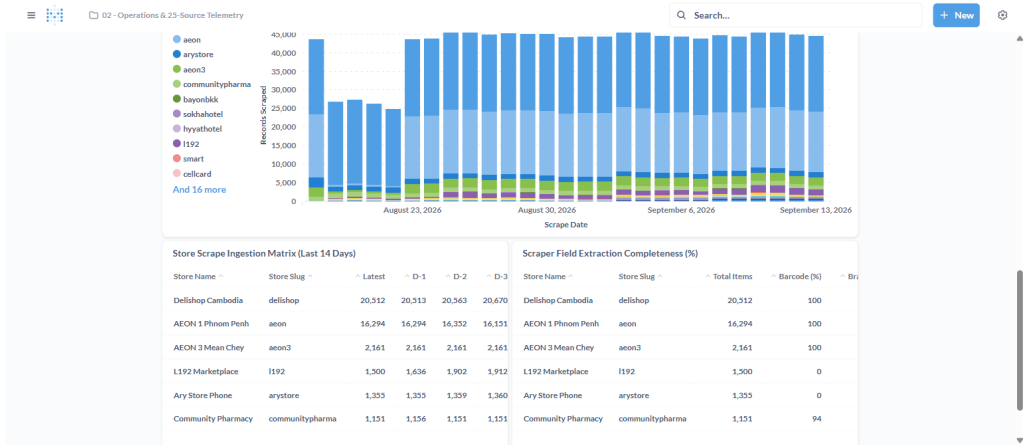


Executive Decision Dashboards in Metabase (Port 3001)



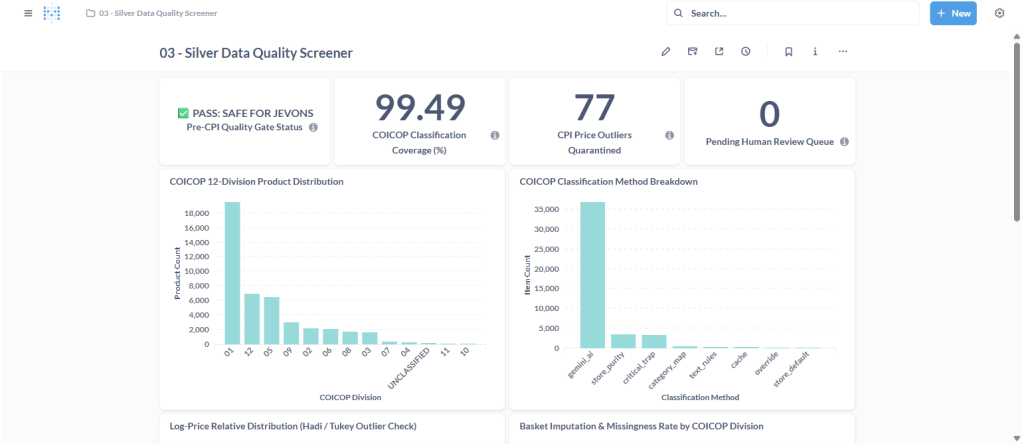
Dashboard 01 (Macro CPI & Inflation Analytics): Real-time Headline/Core CPI tickers, 12 COICOP division table, nowcast convergence band, MLOps model card, and official NIS tracking error indicators.

Operations & 25-Source Telemetry Dashboard



Dashboard 02 (Operations Telemetry): 25-source daily harvest health, extraction success rate (99.3%), latency heatmaps, missing rate tracker, and automated circuit breaker status.

Silver Data Quality Screener Dashboard



Dashboard 03 (Silver Quality Screener): Outlier price shock monitors, MAD filter alerts ($> 3.5 \times \text{MAD}$), Store Priority verification, and AI classification cache memoization stats.

Strategic Policy Roadmaps for National Institutions

National Institute of Statistics (NIS)

- **Axiomatic Jevons:** Adopt geometric mean to eliminate +1.18% formula drift.
- **Hybrid Pipeline:** Run automated scraping alongside field surveys.
- **Scanner Data:** Establish automated POS feeds with major retail chains (AEON, Chip Mong).

Ministry of Economy & Finance (MEF)

- **Early Warning:** Real-time alerts on food/fuel spikes weeks before monthly reports.
- **Dynamic Fiscal Deflators:** Calibrate procurement deflators in real-time.
- **Targeted Safety Nets:** Trigger cash transfers when basic food indices surge.

National Bank of Cambodia (NBC)

- **FX Pass-Through:** Track daily how USD/KHR moves pass into supermarket shelf prices.
- **Refined Core Inflation:** Filter out supply shocks to guide monetary policy.
- **Liquidity Management:** Proactive open market operations during intra-month drift.

Summary of Key Contributions

What This Research Delivered

- ❶ **First Sovereign Daily CPI System:** End-to-end Medallion pipeline ingesting 25 endpoints daily at 08:00 AM ICT across all 12 COICOP divisions in Phnom Penh.
- ❷ **Accurate Entity Matching:** 7 physical specification guards + multilingual 768-dim embeddings preventing false merges and shrinkflation.
- ❸ **Streamlined Classification:** Store Priority routing ($O(1)$) + Google Gemini AI with cache memoization and domain trap guardrails (98.6% Macro F1, 0 mismatches).
- ❹ **Axiomatic Mathematical Integrity:** Jevons geometric mean eliminating +1.18% upward drift.
- ❺ **Sub-Percentage Nowcasting:** Two-Stage Hybrid Ridge Nowcaster with out-of-sample tracking error of only **+0.0181 pp** against official NIS releases.

Academic & System Validation

- **Full Test Suite:** 386/386 unit and integration tests passing (100%).
- **Zero-Defect Warehouse:** 955,789+ rows, 100% classified, 0 code mismatches.
- **Production Telemetry:** 3 live Metabase dashboards (Port 3001).

Digital Public Infrastructure

Establishes a resilient, open, and scalable Digital Public Infrastructure for macroeconomic governance in Cambodia.

Thank You Very Much!

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Questions & Discussion

Floor is now open for questions, comments, and jury feedback.

